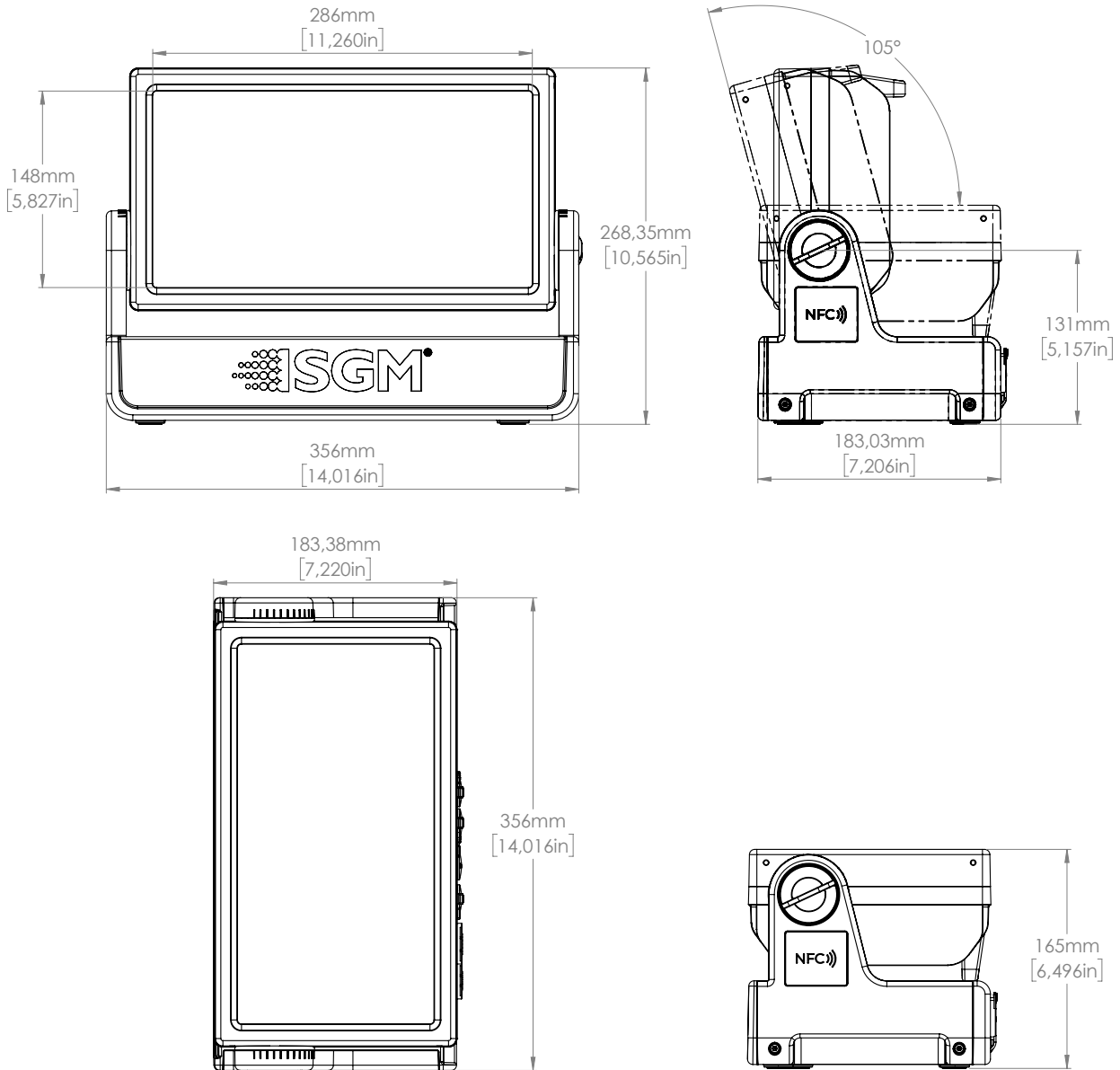


USER MANUAL

PALCO-3 Wash



PALCO-3 WASH DIMENSIONS



All dimensions in millimetres and inches. Drawing not to scale

This manual covers installation, use, and maintenance of the PALCO-3 Wash. A digital version is available at www.sgmlighting.com or upon request via support@sgmlighting.com. The information in this document is subject to change without notice. SGM and all affiliated companies disclaim liability for any injury, damage, direct or indirect loss, consequential or economic loss, or any other loss occasioned by the use of, inability to use, or reliance on the information contained in this manual. The SGM logo, the SGM name, and all other trademarks in this document pertaining to SGM services or SGM products are trademarks owned or licensed by SGM, its affiliates, and subsidiaries. This edition applies to firmware version 1.00 or later.

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WARNING! READ THE FOLLOWING SAFETY PRECAUTIONS CAREFULLY BEFORE UNPACKING, INSTALLING, POWERING OR OPERATING THE DEVICE.



SGM fixtures are intended for professional use only. They are not suitable for residential use.

Les fixtures SGM sont impropres à l'usage domestique. Uniquement à usage professionnel.

This product must be installed in accordance with the applicable installation code by a person familiar with the construction and operation of the product and the hazards involved.

Ce produit doit être installé selon le code d'installation pertinent, par une personne qui connaît bien le produit et son fonctionnement ainsi que les risques inhérents.



DANGER! RISK OF ELECTRIC SHOCK DO NOT OPEN THE DEVICE!

- The SGM fixture shall be serviced only by the manufacturer or an authorized service agent.
- Disconnect power before installing or servicing to avoid electrical shock.
- Ensure that the device is electrically connected to earth (ground).
- Do not apply power if the device or mains cable is in any way damaged.
- Do not immerse the fixture in water or liquid.



WARNING! TAKE MEASURES TO PREVENT BURNS AND FIRE!

- Install in a location that prevents accidental contact with the device.
- Install only in a well-ventilated space.
- Install only in accordance with applicable building codes.
- Do not paint, cover, or modify the device, and do not filter or mask the light.
- Keep all flammable materials well away from the device.

ALLOW THE DEVICE TO COOL FOR AT LEAST 15 MINUTES AFTER OPERATION BEFORE TOUCHING IT. THE EXTERIOR SURFACE TEMPERATURE MAY REACH UP TO 75°C (167°F) AT STEADY STATE.



WARNING! TAKE MEASURES TO PREVENT PERSONAL INJURY. THE LUMINAIRE SHOULD BE POSITIONED SO THAT PROLONGED STARING INTO THE LUMINAIRE AT A DISTANCE CLOSER THAN 5.6 M IS NOT EXPECTED.

- Take precautions when working at height to prevent injury due to falls.
- For Permanent Outdoor Installations (POI), ensure that the fixture is securely fastened to a load-bearing surface with suitable corrosion-resistant hardware.
- For elevated installations, secure the fixture with suitable safety cables, and always comply with relevant load dimensioning, safety standards, and requirements.

Please visit www.sgmlighting.com for the latest version of this user manual / safety information leaflet. Due to continuous improvements, the instructions may change without notice. SGM always recommends the latest available firmware version from www.sgmlighting.com.



VISUAL INSPECTION

All users of the SGM fixtures should regularly clean those parts of the fixture directly exposed to the elements, such as the external housing and front lenses. Additionally, all owners of the SGM fixtures must periodically check the external housing of the fixture for structural breaks, deterioration, cracked lenses, or loose screws. To ensure proper operation, but also to prevent the risk of potential accidents, do not use the fixture if the lens, housing, or power cables are damaged. If parts of the fixture appear to be missing, cease use immediately and contact SGM support.



WIRING

When installing fixtures in a permanent installation, ensure power and data cable leads are installed as a service loop to an appropriately rated junction box using suitable cable strain reliefs/glands. All installed fixtures must be securely mounted, and service loop appropriately protected for installation location. All electrical wiring and connections should be completed by a qualified electrician.

Separation of field installed power limited circuit (dimming/control) wiring from the branch circuit wiring in the outlet box are to be made in accordance with local and/or national electrical installation codes.



SAFETY PRECAUTIONS

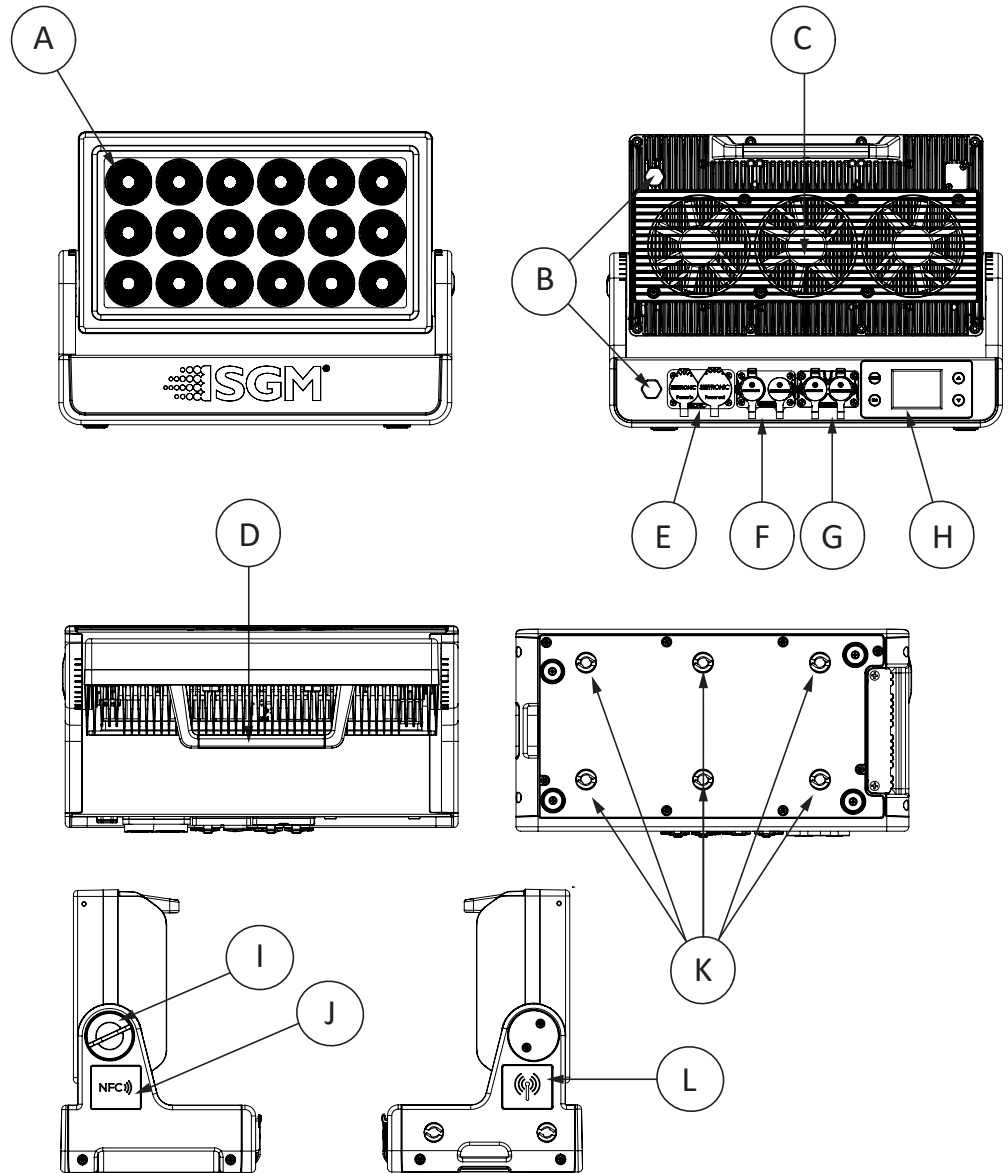
When using electrical equipment, basic safety precautions should always be followed including the following:

- Do not mount near gas or electric heaters.
- Permanently installed equipment should be mounted in locations and at heights where it will not readily be subjected to tampering by unauthorized personnel.
- The use of accessory equipment not recommended by the manufacturer may cause an unsafe condition.
- Do not use this equipment for other than intended use.
- Refer service to qualified personnel or authorized service centres.
- Do not look directly into the beam for long periods of time, when the fixture is on.
- The fixture shall, under no circumstance, be covered with insulating material of any kind.

READ AND FOLLOW ALL SAFETY INSTRUCTIONS.

IDENTIFICATION AND TERMINOLOGY OF PALCO-3 WASH

- A: 18 x RGBL LEDs
- B: GORE-TEX membrane
- C: Cooling fans
- D: Handle
- E: Power In/Out
- F: DMX In/Out
- G: Ethernet In/Out
- H: Control Panel
- I: Tilt Lock
- J: NFC Tag
- K: Holes for Omega bracket
- L: LumenRadio CRMX Antenna



Illustrations might vary from received products. This is subject to change without notice.

UNPACKING

Unpack the device and inspect it to ensure that it has not been damaged during transport. The fixture is shipped with:

- Rubber foam flight case-insert.
- 2 x Omega bracket with 1/4-turn fasteners.
- Safety information leaflet.

APPLICATION CONSIDERATIONS

The standard fixture is IP65-rated and is designed for both indoor and outdoor events. This means that it is protected from:

- Dust, to the degree that dust cannot enter the fixture in sufficient quantities as to interfere with its operation.
- Lower pressure water jets from any direction.

When using standard fixtures outdoors or in wet locations, ensure that:

- The DMX connection out of the last fixture in a chain is properly sealed, in accordance with the ingress protection (IP) requirements.
- The DMX out of the last fixture is terminated with a 120 Ohm resistor between pin 2 and 3 (as per RS485 standards).

When selecting a location for the device, ensure that:

- It is situated away from public thoroughfares and protected from contact with people.
- It has adequate ventilation.
- It is not immersed or submerged in water.

HANDLING AND TRANSPORTATION

Always use the supplied packaging or suitable flight case for transportation and storage. Never carry the fixture by connected cables or wires.

RIGGING

The fixture may be installed in any orientation, on the ceiling or on a wall surface.

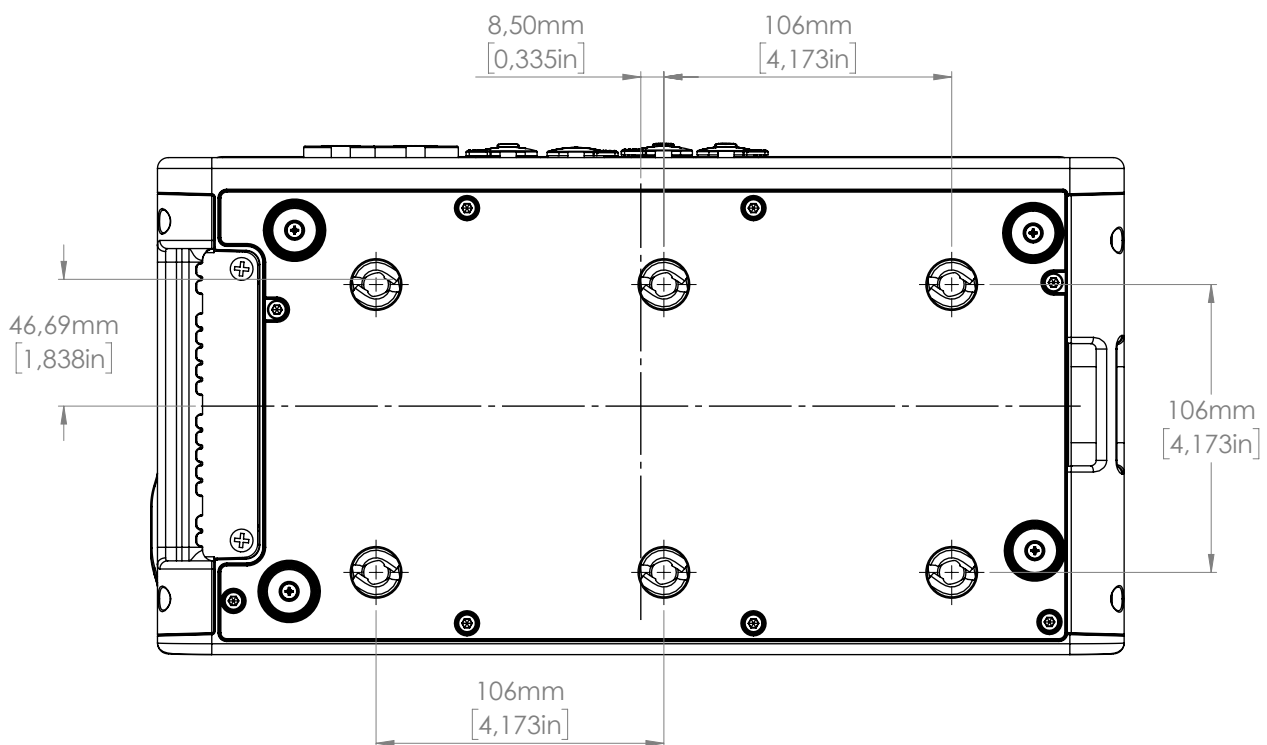
When installed horizontally with a downward beam-angle, water or dirt can potentially pool in the fan wells. Under normal operation the moisture will evaporate.

In locations with high rainfall, consider locating the fixture under cover or an overhang, or modify the position and orientation of the fixture to minimize pooling.

All SGM fixtures have locking points in the base for installation and rigging.

Always use the supplied omega bracket to rig a standard fixture. Lock the bracket with the 1/4-turn fasteners.

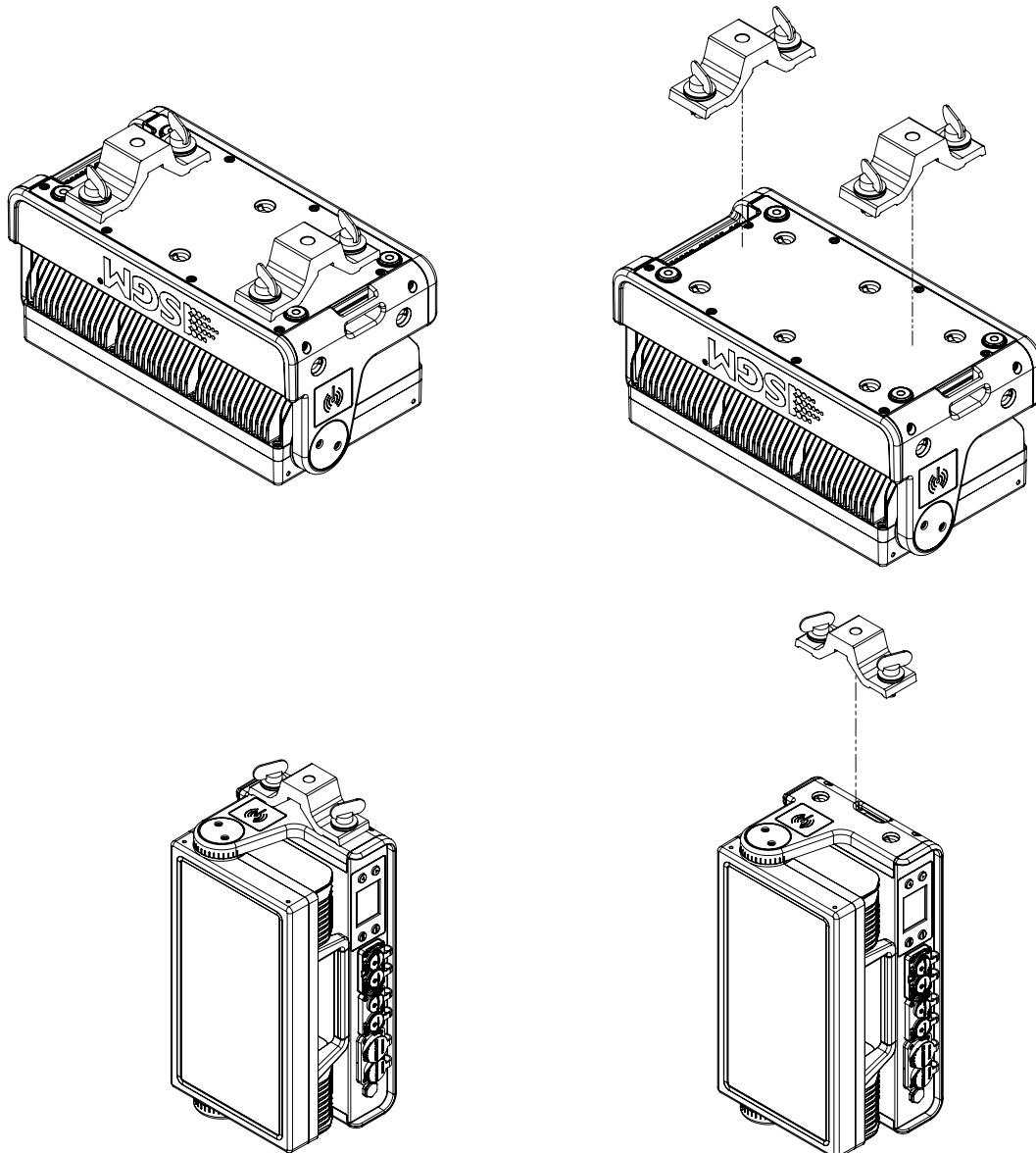
PLEASE NOTE! 1/4-TURN FASTENERS ARE ONLY LOCKED WHEN TURNED FULLY CLOCKWISE. DEPENDING ON THE STRUCTURE, PLEASE USE APPROPRIATE AND SECURE METHODS FOR MOUNTING THE OMEGA BRACKETS.



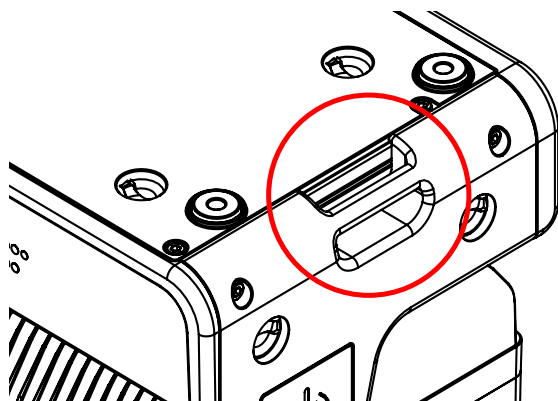
OMEGA BRACKET

Start the standard rigging process by blocking the lower working area, and make sure the work is performed from a stable platform.

1. Check that the clamp/bracket is undamaged. Supporting structure should be capable of bearing at least 10 times the weight of all installed fixtures, lamps, cables etc.
2. Bolt a clamp or bracket securely to the omega bracket with a M12 bolt (min. grade 8.8) and a lock nut. For longer term installations, choose outdoor rated hardware.
3. Align the omega bracket with the fixture base. For standard Omega Bracket, insert the fasteners into the fixture base bracket, and turn both levers a full 1/4 turn clockwise to lock.
4. Hang the fixture on a truss or other structure. Tighten the clamp/bracket.
5. Verify that there are no combustible materials, cables, or surfaces to be illuminated within 0.3 m (12 in.) of the fixture.
6. Check that there is no risk of the head/yoke colliding with other fixtures or structures.



SAFETY WIRE



Fasten a safety wire (not shown) between the load-bearing support structure and the safety wire attachment point on the device.

The safety cable (not included in the package) must:

- Bear at least 10 times the weight of the device (SWL).
- Have a minimum gauge of 4 mm.
- Have a maximum length (free fall) = 30 cm (12 in.).



WARNING! ALWAYS SECURE AN ELEVATED FIXTURE WITH A SAFETY WIRE. ALWAYS USE A SAFETY WIRE OF A GRADE AISI 316 STEEL. MAKE SURE THE SLACK OF THE SAFETY WIRE IS AT A MINIMUM. NEVER USE THE YOKE/HANDLE FOR SECONDARY ATTACHMENT.

TILT LOCK

The fixture head can pivot through 0-105°.

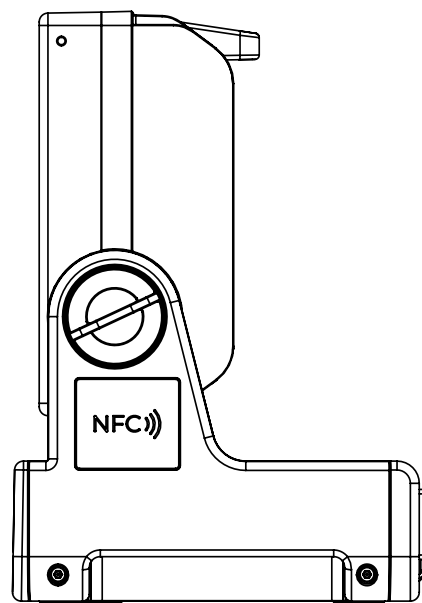
To adjust the tilt angle:

- Loosen the rotary lock by turning it counter-clockwise.
- Tilt the fixture head to the desired position and hold it.
- Lock the position by re-tightening the tilt lock clockwise

Apply reverse procedure before storing the fixture into the flight case insert.

CAUTION!

ALWAYS LEAVE A FIXTURE TO COOL OFF FOR 15 MINUTES AFTER OPERATION BEFORE HANDLING.



POWER

The power input required is 100–240 VAC, 50/ 60 Hz AC mains power supply, and draws a maximum of 400 Watts. Power is connected via a “Powercon” type connector. Power plug is as per local code.

Wire	Color	Symbol	Conductor
	Black	L	live
	White	N	neutral (L2)
	green/yellow	$\perp$ or $\oplus$	ground (earth)

- Connect the black wire to live
- Connect the white wire to neutral
- Connect the green/yellow wire to ground (earth)

The power cable must be fitted with a grounded connector intended for exterior use.

PLEASE NOTE! PROTECTIVE CAPS MUST BE SECURELY MOUNTED ON ANY UNUSED CONNECTORS, IN ORDER TO MAINTAIN INGRESS PROTECTION (IP RATING).

CAUTION! DO NOT CONNECT THE FIXTURE TO AN ELECTRICAL DIMMER SYSTEM, AS DOING SO MAY CAUSE DAMAGE

DATA

The fixture is controllable using a DMX, sACN or Art-Net control device, and can be connected using either a DMX cable, Ethercon style cable, or via the fixture's built-in CRMX wireless receiver system.

DMX

When using a cabled DMX system, connect the DMX-In cable to the input connector and DMX-Out cable to the output. Both are located on the rear of the fixture's base (chassis mounted male and female 5-pin XLR plugs).

For outdoor installations, use only IP65-rated XLR connectors suitable for outdoor use. Terminate the DMX out cable of the last fixture in the data link with a 120-ohm DMX termination.

PLEASE NOTE! SGM FIXTURES PROVIDE A PASSIVE DMX THRU SIGNAL AS DMX OUT, INSTEAD OF AN ACTIVE OUTPUT SIGNAL.

ETHERCON

Using sACN or Art-Net is done through the ethernet connectors. These are compatible with all Ethercon or RJ45 style connectors. The 2 ethernet ports can serve in either a daisy chain topology or can be fed directly with the ethernet lines, providing a double input for ethernet protocols. For outdoor installations, use only IP65-rated XLR connectors suitable for outdoor use. In a setup where the ethernet ports are in a daisy chain, the fixture will still pass through the ethernet signal in the event the fixture loses power.

CONNECTING A WIRELESS TRANSMITTER

The fixture is designed to look for wireless transmitters in 'connect' state when this option is not yet enabled. The fixture comes tested for wireless functionality from the factory, therefore the beginning of the pairing process is disconnect from the factory wireless DMX transceiver.

To connect the fixture to a wireless transmitter:

1. Log off the currently paired wireless transmitter. Go to SETTINGS → WIRELESS DMX → LOG OFF in the menu. Fixture confirms logged off.
2. Initialize connection on the wireless transmitter.
3. Verify on the display that a wireless symbol will appear in the top right corner.

SIGNAL PRIORITY

The fixture can be paired to an active wireless transmitter while also being connected to a cabled DMX or ethernet signal. By default, the fixture will prioritize cabled DMX signal. If there is not cabled DMX signal, it will use the ethernet signal. If it is also not receiving any ethernet signal, it will switch to wireless DMX.

Through the Settings menu in the display or RDM, this can be changed so that the fixture will prioritize wireless DMX over Cabled DMX and ethernet.

The active input type is displayed in the top right corner of the display. The wireless signal strength can be checked from the System menu in the display or via RDM by using an external RDM device.

NFC

The fixture includes a built-in Near Field Communication (NFC) interface, providing compatibility with both Android and iPhone devices for basic configuration and information access. The SGM NFC app can be downloaded from the App Store or Google Play Store.

The NFC tag is positioned on the side of the fixture. To access the configuration, open the SGM NFC app and hold your smartphone as close to the tag as possible. Note that the fixture does not need to be powered for the NFC functionality to operate, and the NFC tag remains accessible even when the fixture is stored in the flight case.

USER INTERFACE

The fixture can be set up by using the control panel and the multi-line display on fixture's base or through RDM.

The display shows the current status and menu of the fixture. It is used to configure individual fixture settings and read error messages. Before turning on the fixture, make sure the power cable is properly connected. When the fixture is powered on it will boot and reset before displaying the selected operating mode and DMX start address. Navigate through the menus and options using the arrow buttons, and select items using the ENTER button.

USING THE DISPLAY PANEL

- Press the 'ENTER' button to access the menu or make a selection.
- Press the 'ESC' button to take a step back in the menu.
- Press the arrow buttons to scroll up and down in the menus.



DISPLAY

FIXTURE UNIVERSE (A)

Displays the current fixture universe. The universe is altered directly from this view.

DMX ADDRESS (B)

Displays the current DMX address. The DMX address is altered directly from this view.

EXTERNAL DATA INDICATOR (C)

The DMX signal indicator will flash when the DMX signal is received. If a wireless transmitter is connected, the wireless signal strength is also displayed.

EXTERNAL DATA PROTOCOL (D)

Shows the external data protocol.

- When 'DMX' is displayed: the fixture responds to data received through cabled DMX.
- When 'CRMX' is displayed: the fixture responds to data received through wireless DMX.
- When 'ART' is displayed: the fixture responds to data received through Art-Net.
- When 'sACN' is displayed: the fixture responds to data received through sACN.

INFORMATION PANEL (E)

The information panel is accessed by pressing ESC. It functions as a way to quickly get an overview of errors, notifications and important settings.

OPERATIONAL MODE (F)

Displays the current DMX mode.

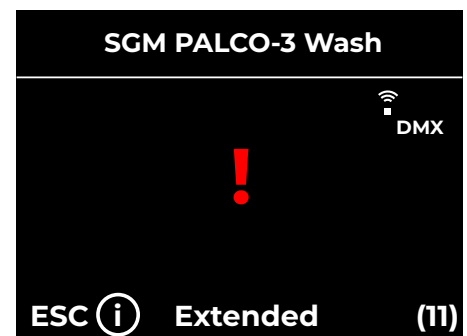
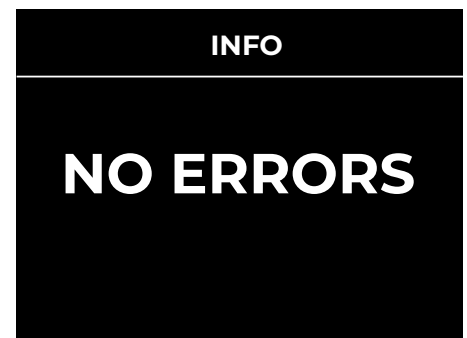
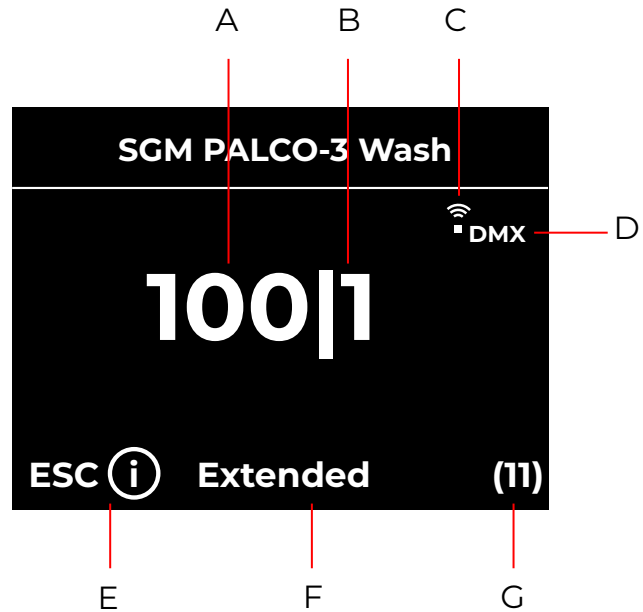
NEXT AVAILABLE ADDRESS (G)

The next available DMX address will show immediately, depending on the fixture's DMX footprint.

WARNING INDICATOR

If any errors are detected, the exclamation point will alternate with the current display for easy detection.

To read the error message, do to the information panel by pressing ESC.



CONFIGURATION

The PALCO-3 fixture is configured using either the display interface or SGM software; the SGM RDM Addressing Tool. This software is Windows® PC based and connects to the fixtures via DMX512/RDM using the SGM USB uploader cable.

ABOUT DMX

The fixture can be controlled using signals sent by a DMX controller on a number of DMX channels.

DMX is the USITT DMX512-A standard, based on the RS-485 standard. The signal is sent as DMX data from a console (or a controller) to the fixtures via a shielded twisted pair cable designed for RS-485 devices.

The cables can be daisy chained between the fixtures, and up to 32 fixtures can be connected on the same DMX link. Up to 300 m. (~1000ft.) of cable is achievable with high quality DMX cables. All DMX links must be terminated by connecting a DMX termination plug to the last fixture's 5 pin DMX out connector.

PLEASE NOTE! STANDARD MICROPHONE CABLE IS NOT SUITABLE FOR TRANSMITTING DMX. UP TO 32 FIXTURES CAN BE LINKED TO THE SAME DMX CHAIN. ADDITIONAL FIXTURES WILL OVERLOAD THE LINK.

DMX START ADDRESS

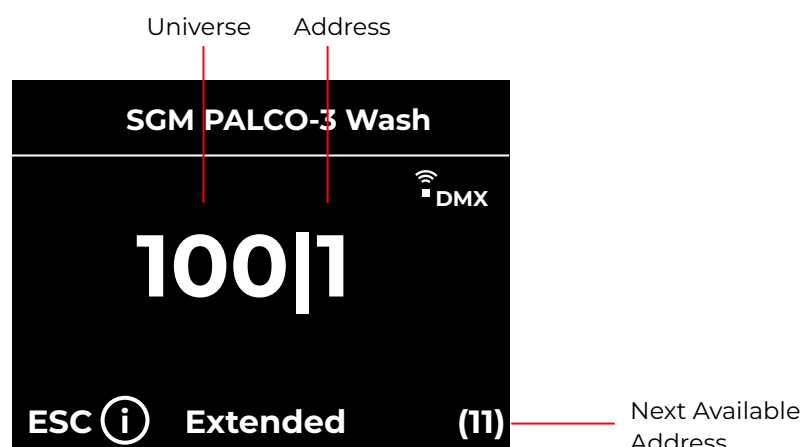
The fixture can be operated in different DMX modes. For any of the modes, the first channel used to receive data from a DMX control device is known as the DMX start address.

For independent control, each fixture must be assigned to its own DMX start address. For example, if the first RGBL fixture is set to STANDARD mode with a start DMX address of 101, the following RGBL fixture in the DMX chain should then be set to a DMX address of 111. As the first fixture uses all the first 10 DMX channels, including channel 101, the next available channel is 111 (101+10=111).

If two or more fixtures have the same DMX start address, they will behave identically. Incorrect settings will result in unpredictable responses from the lighting controller. Address sharing can be useful for diagnostic purposes and symmetrical control.

SETTING UNIVERSE AND ADDRESS

The fixture universe and address are shown on the displays main menu. To change the universe and address, start by pressing the up or down button once. Use the up and down button to first change the universe. When the desired universe is shown press ENTER to start changing the address by using the up and down buttons. Once the correct address is shown press ENTER to store the universe and DMX address. Note that if the fixture is controlled through cabled/wireless DMX, the universe setting has no effect. The universe is only used if the fixture is controlled via Art-Net or sACN using the ethernet port. For your convenience, the next available DMX address is displayed in the bottom right corner of the display. The fixture also offers the option to set the universe and address through RDM.



SETTING DMX MODE

A menu layer has been implemented to allow selection of the desired DMX mode. To access the available modes:

1. Press [ENTER]
2. "DMX MODES" is displayed, Press [ENTER]
3. Press [Up] or [Down] to select Mode.
4. Press [ENTER]

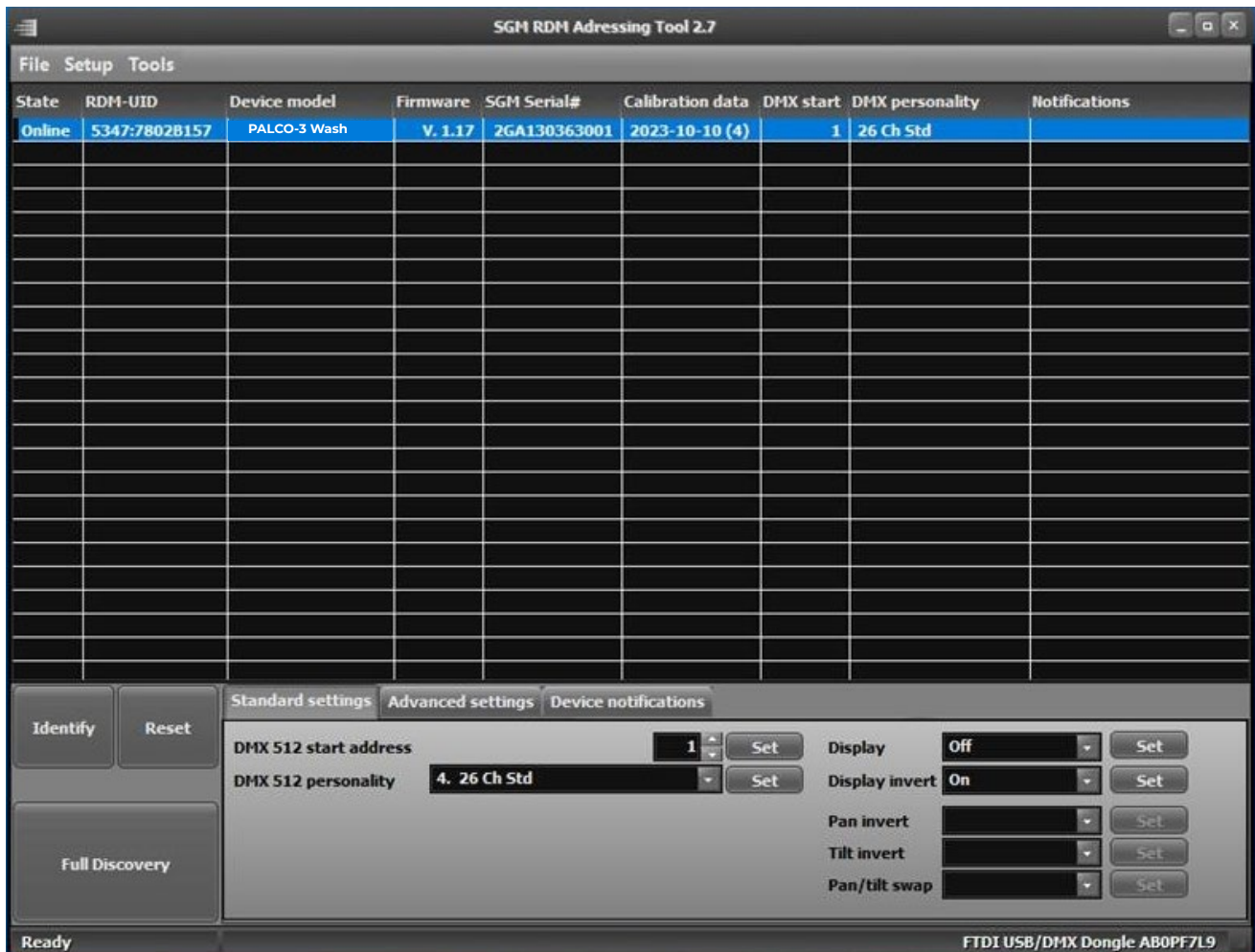
PLEASE NOTE: DMX CHARTS FOR ALL MODES ARE AVAILABLE FOR DOWNLOAD AT WWW.SGMLIGHTING.COM OR BY EMAILING SUPPORT AT SUPPORT@SGMLIGHT.COM

MANUAL OPERATION

The standard fixture can be configured to display a predefined and static color. To set up a static color select ENTER → MANUAL → QUICK COLOR.

1. Quick color is activated as start-up. Select ENTER → SETTINGS → STARTUP MODE → QUICK COLOR.
2. Quick color is activated as signal loss. Select ENTER → SETTINGS → DMX LOSS → QUICK COLOR.

All configuration of fixtures can be done remotely through the SGM Addressing Tool. This software is a windows-based program designed to allow the user to configure the fixture through the RDM protocol.



STEP 1: Launch SGM RDM Addressing Tool. Click Full Discovery and look for the fixture to show in the table.

STEP 2: Select the standard settings tab and enter a DMX address and a personality/mode of operation. These modes must match the control mode in the controller which will control the fixture. Visit www.sgmlighting.com to see all DMX charts available under the respective product, or upon request via support@sgmlighting.com.

Refer to "Factory default" section for all default configuration.

INDIVIDUAL FIXTURE SETTINGS

Display flipping

If the fixture is installed hanging upside down, it might be useful to flip the display so that it is easier to read.

To flip the display, press ENTER and select SETTINGS → FLIP DISPLAY.

Display saver

By default, the display turns off after a short period when the display is not in use. This is ideal when a pitch-black environment is required. Pressing any key will always turn on the display

The display can also be set to always be on. To configure the display settings, press ENTER → SETTINGS → DISPLAY SAVER.

PLEASE NOTE! TO AVOID THE RISK OF DISPLAY DETERIORATION CAUSED BY LONG TERM USE, IT IS RECOMMENDED TO SELECT THE SETTING → DISPLAY SAVER → OFF.

Setting the fan mode

For operating environments where low-noise is a requirement, or where the fixture will be operating at high temperatures, it is possible to adjust the default fan speed by going to SETTINGS → FAN MODE in the menu.

FACTORY DEFAULT

DMX ADDRESS	1
UNIVERSE	0
DMX LOSS	DMX hold
FAN MODE	Auto
DISPLAY SAVER	Display off
FLIP DISPLAY	Normal orientation
QUICK COLOR	Reset
SOURCE PRIORITY	Wired DMX
IP ADDRESS	Default address

SUPPORTED RDM FUNCTIONS

RDM (Remote Device Management) is a protocol enhancement to USITT DMX512 that allows bi-directional communication between the fixtures and the controller over a standard DMX line. This protocol will allow configuration, status monitoring, and management. A RDM controller is needed to get control over the supported parameters. See the tables below for supported RDM functions.

PLEASE NOTE! THE RDM CONTROLLER COMMUNICATES WITH THE FIXTURES TO SHOW ONLY THE AVAILABLE OPTIONS FOR EACH RDM FUNCTION. TABLES ARE SUBJECT TO CHANGE WITHOUT NOTICE.

SENSORS

RDM enables various sensor readouts for remote device monitoring. See the table below for sensors and sensor types.

PLEASE NOTE! THE RDM CONTROLLER COMMUNICATES WITH THE FIXTURES TO SHOW ONLY THE AVAILABLE SENSORS FOR THIS FIXTURE. THE TABLE IS SUBJECT TO CHANGE WITHOUT NOTICE.

NAME	SENSOR TYPE
Led Temperature 1	Temperature
Led Temperature 2	Temperature
Led Temperature 3	Temperature
PCB Temperature	Temperature
PSU Temperature	Temperature

NAME	SENSOR TYPE
Fan RPM 1	Other
Fan RPM 2	Other
Fan RPM 3	Other
Base Fan RPM 1	Other

PID COMMANDS

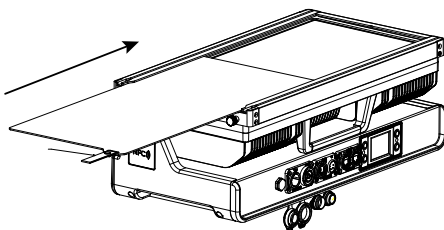
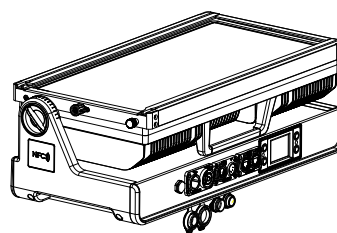
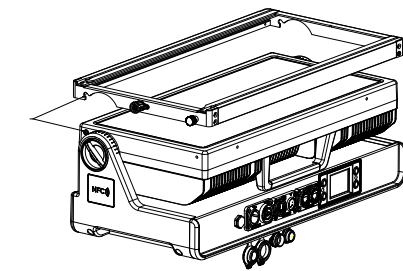
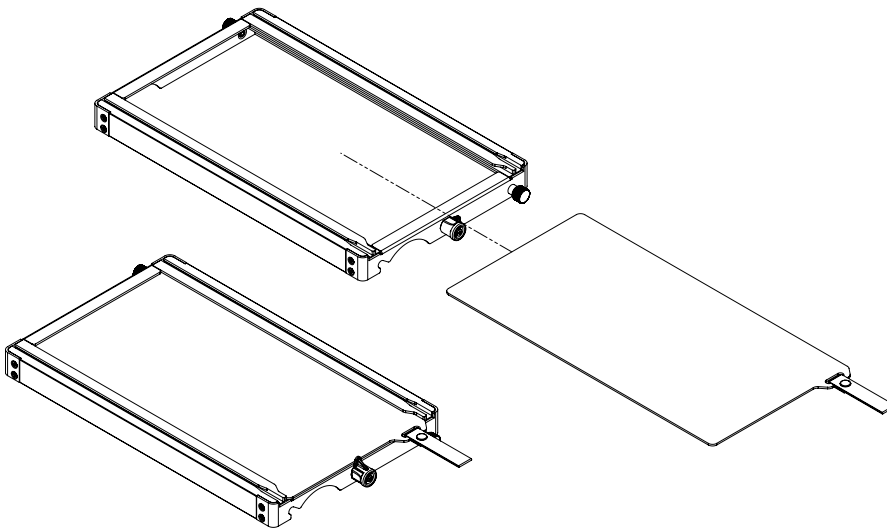
PID	ACTIONS ALLOWED	NAME
0x0060	Get	Device Info
0x00F0	Get/Set	DMX Start Address
0x0081	Get	Manufacturer Label
0x00C0	Get	Software Version Label
0x1000	Get/Set	Identify Device
0x00E0	Get/Set	DMX Personality
0x00E1	Get	DMX Personality Description
0x0080	Get	Device Model Description
0x0051	Get	Parameter Description
0x0200	Get	Sensor Definition
0x0201	Get	Sensor Value
0x0120	Get	Slot Info
0x0121	Get	Slot Description
0x0082	Get/Set	Device Label
0x0401	Get	Lamp Hours

PID	ACTIONS ALLOWED	NAME
0x0400	Get	Device Hours
0x1001	Set	Reset Device
0x0090	Set	Factory Defaults
0x0500	Get/Set	Display Invert
0x0501	Get/Set	Display Level
0x8060	Get	Serial Number
0x8649	Get/Set	Universe
0x8634	Get/Set	DMX Loss
0x863B	Get/Set	Source Priority
0x8625	Get/Set	Fan Mode
0x8647	Get/Set	IP Address
0x8626	Get/Set	CRMX Log Off
0x8635	Get/Set	Start-up Mode
0x8636	Get	Active Error

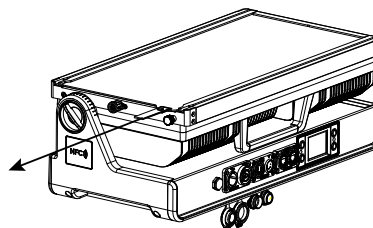
FILTER FRAME

Universal frame mounts onto the fixture

- Filter slides into the frame
- Filter can be exchanged whilst barndoor is mounted
- Frame and filters can stay mounted on fixture whilst fixture is in flight case
- Filters can be exchanged whilst fixture (with mounted frame) is inside flight case



(P1)

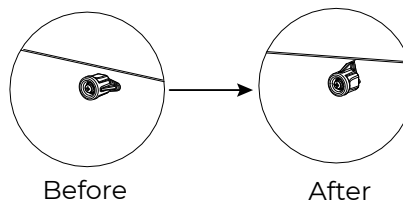
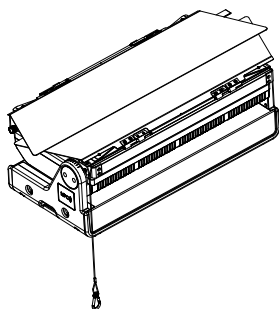
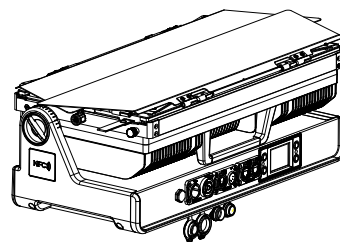
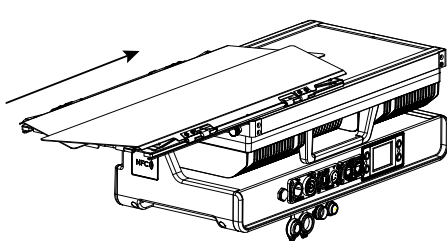
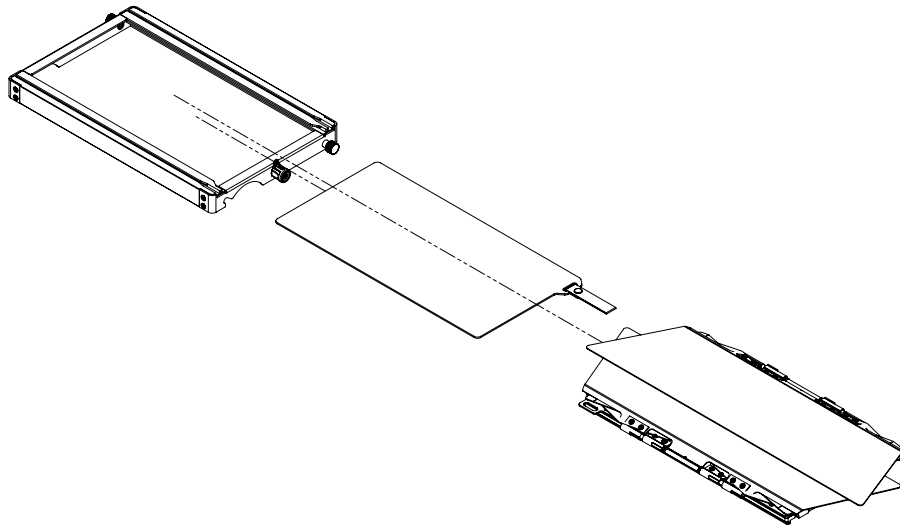


(P2)

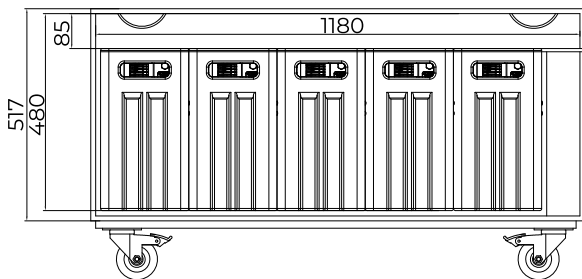
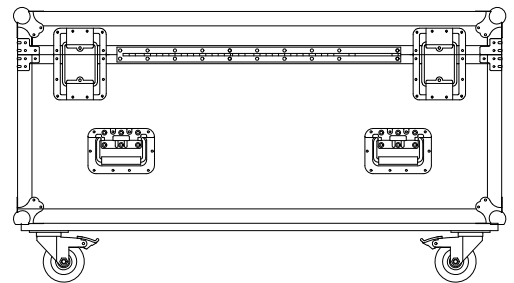
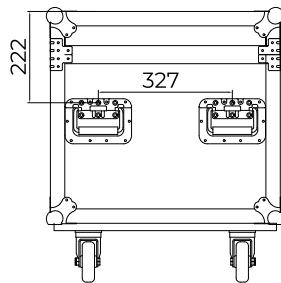
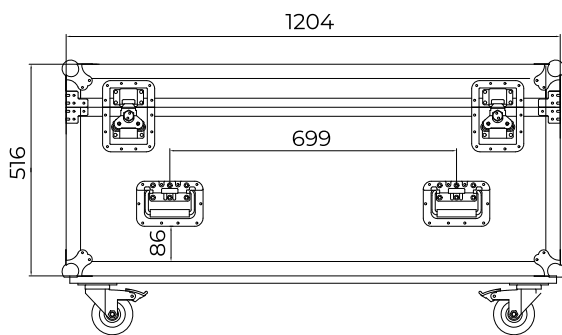
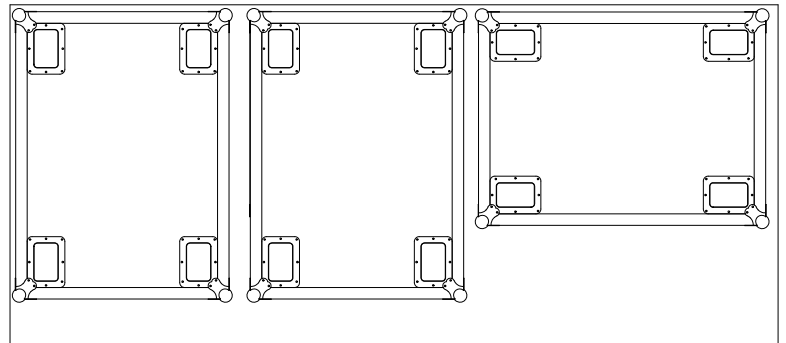
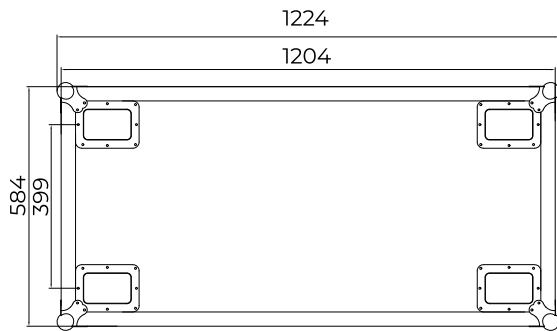
BARNDOOR MOUNTING

Universal frame mounts onto the fixture

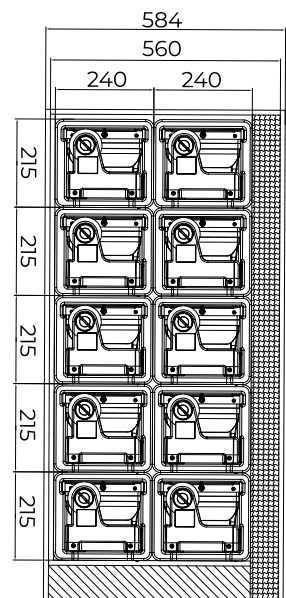
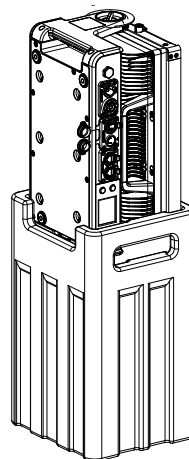
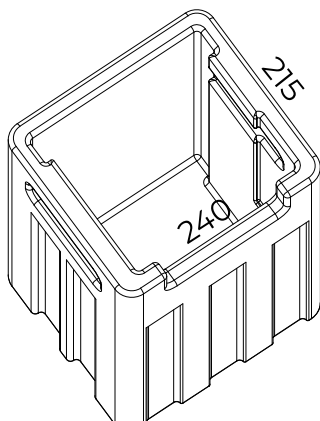
- Barndoor slides in on top of the frame
- Filter can be exchanged whilst barndoor is mounted
- Barndoor needs to be removed before placing fixture into flight case
- Filters can be exchanged whilst fixture (with mounted frame) is inside flight case



PACKING DIMENSIONS

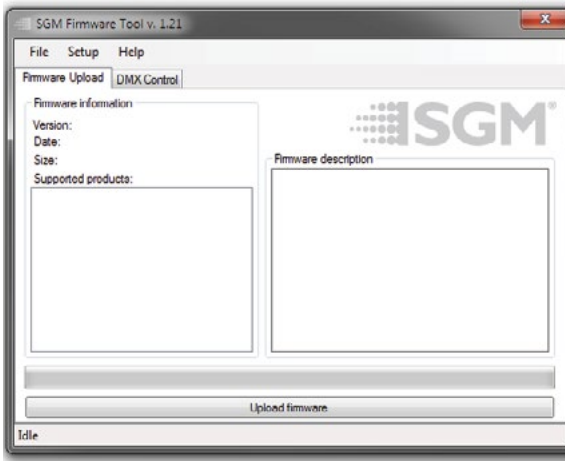


240 x 215 x 395 mm



FIRMWARE UPDATES

The latest firmware, manuals and tools are all available for free download at www.sgmlighting.com. The SGM Firmware Tool is used to update firmware:



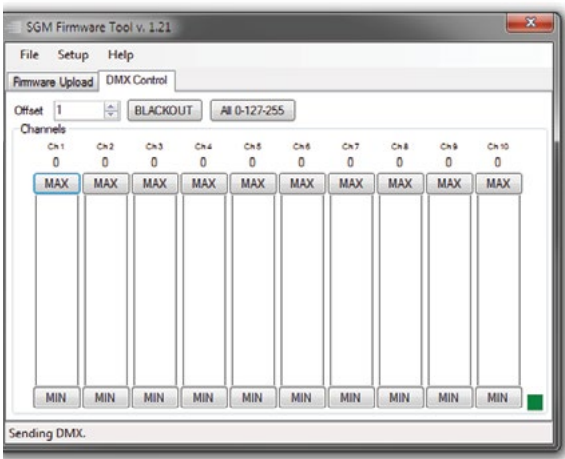
Fixture firmware can be identified in two different ways:

Display

- When powering on a standard fixture, the display shows the current installed firmware version.
- Go to MENU → SYSTEM → FIRMWARE VERSION (on stand-ard).
- Through RDM

RDM

- Connect either Std or POI uploader cable depending on which is more suitable for the installation
- Launch the SGM RDM Addressing Tool
- Click “Full Discovery” fixture should show in the chart. Firmware will be displayed in the firmware column

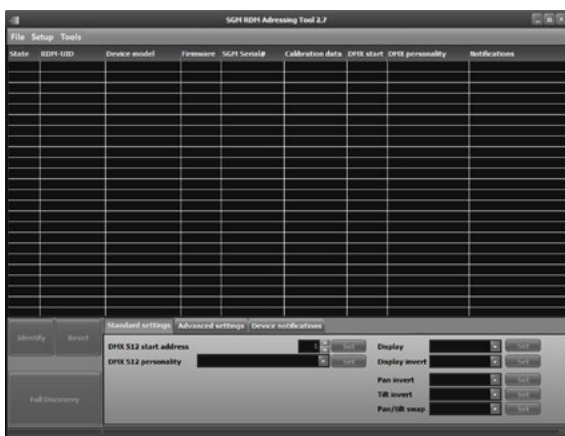


To update a fixture with the latest firmware, use an SGM USB 5-Pin-XLR uploader cable, and a Windows-based computer with the SGM Firmware Tool software installed (available for download at www.sgmlighting.com).

- Download the SGM Firmware Tool software from the SGM website
- Download latest firmware file from product web page on the SGM website
- Connect either Std or POI uploader cable depending on which is more suitable for the installation
- Launch SGM Firmware Tool on a windows PC
- Click “File” then “Open” and navigate to the firmware file needed, select it and click “Open”
- The firmware is now loaded in the uploader, click “Upload Firmware”

The fixture will now update and reboot

The Firmware Tool software offers a simple DMX controller featuring 512 DMX channels for test purposes.



IT IS RECOMMENDED KEEP FIRMWARE UPDATED. THE LATEST FIRMWARE VERSION IS ALWAYS AVAILABLE FOR DOWNLOAD UNDER THE RESPECTIVE PRODUCT AT WWW.SGMLIGHTING.COM.



SGM VACUUM TEST KIT

The Vacuum Test Kit is an accessory suitable for all SGM IP-rated fixtures, made for testing the IP integrity upon reassembly.

In order to ensure the IP-rating of the fixture, it's highly recommended that the fixture is always vacuum tested after installing or swapping any part that might compromise the IP-rating, e.g., swapping the front lens.

SGM disclaims liability for any damage occasioned by the non-use, or inability to use, the vacuum test kit after reassembling the fixture.

USB - XLR UPLOADER CABLE

The SGM USB to DMX cable is an accessory used mainly to update the fixture with the latest SGM firmware. See below how to update the fixture with the latest firmware.

The SGM Uploader cable is also used for controlling the DMX values channel by channel through the Firmware Uploader Tool software (available for download at www.sgmlighting.com).

CLEANING

SGM fixtures with IP65 or IP66-rating do not need any cleaning procedures inside the fixture. However, cleaning the front lens may be needed to achieve the maximum light output after exposure to dust, sand, or dirt. The exterior housing can also be cleaned to get a better look. To maintain adequate cooling, fans must be cleaned periodically.

Whenever necessary, clean the fixture using a soft cloth dampened with a solution of water and a mild detergent. Do not use products that contain solvents, abrasives, or caustic agents for cleaning, as they can cause damage to hardware, cables, and connectors.

The level of cleaning required will vary greatly depending on the operating environment and installation. Therefore, it is recommended to do frequent check-ups the first few weeks of operation to see how often cleaning is necessary.

TROUBLESHOOTING

PROBLEM	POTENTIAL CAUSE(S)	REMEDIES
Fixture does not respond or appears to be completely dead.	No power to the fixture.	Confirm that the power is switched on, confirm that the cables are plugged in.
	Main fuse is blown.	Contact SGM support or certified SGM service partner.
Fixture suddenly turned off.	Power was turned off.	Check the power supply, switches and breakers.
Fixture suddenly stopped responding.	Wireless transmitter or connections was disconnected/ tampered with.	Inspect the wireless transmitter and connections.
	DMX cables were disconnected.	Inspect DMX cables.
Fixture operates irregularly/ abnormally.	DMX cable polarization is inverted (pin 2 + 3).	Install a phase-inverter or replace cables.
	DMX link is not terminated.	Install a XLR 120 ohm DMX termination at the end of the DMX link.
	Corrupted DMX cable.	Replace or repair defective cables and/ or connectors.
	A corrupted fixture generates noise/ disruptions on the DMX link.	Track and isolate the corrupted fixture.
Color is uneven.	The minimum values are out of calibration.	Contact your local SGM dealer or support@sgmlight.
	The SGM Calibration Data set has been lost.	Contact your local SGM dealer or support@sgmlight.

The fixture can be used with a variety of accessories.
Contact your local SGM dealer to get the latest pricing and news about available accessories.

PLEASE NOTE! THE LIST BELOW IS SUBJECT TO CHANGE WITHOUT NOTICE.

ORDERING INFORMATION

The ordering information of the PALCO-3 Wash and it's variants can be found on the [PALCO-3 Wash product page](#)

ACCESSORIES

Barndoor 8-way, PALCO-3	P/N: 83063124
Universal Frame, PALCO-3	P/N: 83063127
Barndoor & Universal Frame Bundle, PALCO-3	P/N: 83063130
Filter Set & Universal Frame Bundle, PALCO-3.....	P/N: 83063133
Wall Wash Filter, PALCO-3	P/N: 83063136
Scrim Filter, PALCO-3	P/N: 83063139

APPROVALS AND CERTIFICATIONS

Conforms to 2014/35/EU: Low Voltage Directive
Conforms to 2014/30/EU: EMC Directive
Conforms to 2011/65/EU: RoHS2 Directive
Conforms to UK SI 2016 No. 1101: The Electric Equipment (Safety) Regulations 2016
Conforms to UK SI 2016 No. 1091: Electromagnetic Compatibility Regulations 2016
Conforms toUK SI 2012 No. 3032: Restriction of the Use of Certain Hazardous Substances
in Electrical and Electronic Equipment Regulations 2012 (RoHS2)
Conforms toUL Std. 1573
Certified toCSA Std. C22.2 No.166



- This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.
- Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.
- Please note that changes or modifications of this product is not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

The information in this document is subject to change without notice.
For the latest information, visit www.sgmlighting.com.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



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